

Software Requirements Specification (SRS)

Project: Lost Items Recovery Platform

Introduction

The Lost Items Recovery Platform is a web and mobile application designed to help users report lost and found items, connect with each other, and facilitate the recovery process. This document details the software requirements, features, and operations of the platform.

Project Scope

The platform allows users to report lost or found items, search for these items, and communicate with the relevant parties (loser or finder). The platform will offer a user-friendly experience on both web and mobile applications.

Functional Requirements

3.1 User Registration

- Users must provide basic details (name, email, password, location).

3.2 Ad Posting

- Users can post ads for lost or found items with the following details:
 - **Item Type**: Options include money, keys, wallet, cards, tools, pets, etc.
 - **Location**: Specify exact address, public place, or transportation type (with route).
 - **Date and Time**: Indicate the time of loss or finding.
 - **Comments Section**: Add any additional details about the item.

- **Image Upload**: Option to upload a photo of the item.

3.3 Search Functionality

- Users can search for ads using filters:
 - **Item Type**.
 - **Location**.
 - **Date and Time**.
 - **Additional Details**.
- **Image Search**: Users can upload an image to match it with similar ads.

3.4 Communication (Chat Room)

- Users can initiate communication with the ad publisher (loser or finder) through a chat room.
- **Chat Notifications**: Users will receive notifications when someone wants to talk.
- **End of Conversation**: After the chat, users can rate each other.

3.5 Rating System

- Users will rate each other on a scale of 1-5 stars after the conversation, ensuring trust and reliability.

4. Non-Functional Requirements

4.1 Performance

- The platform should handle a large number of users and data requests efficiently.

4.2 Security

- Ensure user data protection, including secure login, encrypted communications, and data storage.

4.3 Usability

- The platform should be easy to navigate for all types of users.
- Mobile app should be intuitive and responsive.

4.4 Compatibility

- The platform will be compatible with modern web browsers and mobile operating systems (iOS/Android).

5. System Features

5.1 Web Application

- **Home Page**: Features buttons for "Lost Something" and "Found Something."
- **Ad Posting**: Allows users to quickly post ads.
- **Search Bar**: For easy access to search functionalities.
- **Notification Icon**: Displays new alerts.
- **User Account Management**:
 - **Chat History**: Displays past conversations.
 - **History**: Displays past ad interactions.

5.2 Mobile Application

- **Home Tab**: Displays the most recent ads.
- **Location Access**: Users are prompted to share their location.
- **Ad Posting**: Includes options for posting lost or found items.
- **User Account Management**:
 - **Chat History**: Shows past conversations.
 - **Transaction History**: Shows past ad interactions.

- **Notification Feature**: Alerts users to new ads in their area.

6. User Interfaces

6.1 Web Interface

- Home page with two main options: Lost Something or Found Something.
- Filters for searching ads.
- Chat window for communication.

6.2 Mobile Interface

- Display recent ads in the home tab.
- Filter and search functionalities for lost or found items.
- Chat functionality for user interaction.

7. Technical Requirements

- **Frontend**: HTML, CSS, JavaScript (React for web, Flutter for mobile).
- **Backend**: Node.js, Express for RESTful APIs.
- **Database**: MongoDB or SQL for data storage.
- **AI Integration**: For image-based searches (optional).

9. Timeline

Phase 1: UI Design (Deadline: Feb 10)

- Design the web and mobile UI.
- Create wireframes and high-fidelity designs.

Phase 2: Front-End and Flutter Development (From 10 Feb –

Deadline: Apr 10)

- Implement the designs and integrate with back-end services.

Phase 3: Back-End Development and AI Integration (From Apr 10 –

Deadline: July 2024)

- Develop APIs, database schema, and integrate AI-based search.

Note

It is expected that some work periods will overlap with exam periods during the following times:

- From mid-December to the end of January.
- Last week of March.
- From early May to mid-June.